

Architecture Selection – Public Bus Live Tracking & Crowding Estimator

Architecture Selected: Microservices Architecture

Architectural Choice: We chose Microservices Architecture for the Public Bus Live Tracking & Crowding Estimator system.

Reason 1 – Scalability: The system receives GPS telemetry from up to 1,000 active buses, with GPS coordinates being ingested every 5 seconds. Separating GPS ingestion, ETA calculation and crowding estimation into independent services allows each service to handle its workload and scale independently.

Reason 2 – Fault Isolation: The system performs multiple functions including GPS ingestion, ETA calculation and passenger crowding estimation. With microservices, failures in one service can be isolated from other services, improving the reliability of the commuter application and fleet-controller dashboard.

Security Advantage: Each service can use controlled APIs with authentication and authorization. Public commuter services can be separated from fleet-controller functions, reducing unauthorized access to operational information.

Performance Benefit: GPS telemetry can be processed independently by the GPS Data Ingestion Service and passed to the ETA Tracking Engine. This allows high-frequency GPS information to be processed efficiently without requiring the entire application to process every event.

Conclusion: Microservices Architecture is suitable for the Public Bus Live Tracking & Crowding Estimator because it supports scalability, fault isolation, independent processing and efficient handling of high-frequency GPS and passenger data.